

Geometry

Unit 10

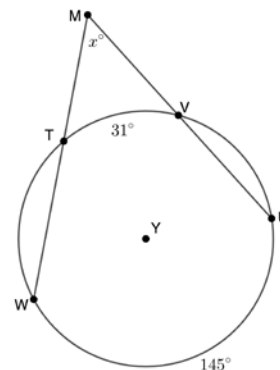
Lesson 8 Homework

Name _____

Date _____

Period _____

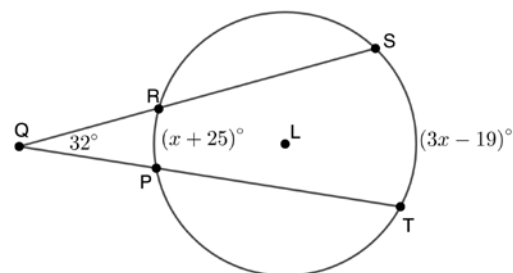
For questions 1-2, Circle Y has points W , T , V , and U on the circle. Secant lines $\overline{WM}$ and $\overline{UM}$ intersect at point M outside the circle. The $m\widehat{UW} = 145^\circ$, $m\widehat{TV} = 31^\circ$, and $m\angle TMV = x^\circ$.



1. Write a formula that can be used to find the value of x .

2. Find the value of x .

For questions 3-5, Circle L has points T , P , R , and S on the circle. Secant lines $\overline{TQ}$ and $\overline{SQ}$ intersect at point Q outside the circle. The $m\widehat{ST} = (3x - 19)^\circ$, $m\widehat{PR} = (x + 25)^\circ$, and $m\angle PQR = 32^\circ$.



3. Find the value of x .

4. Find the $m\widehat{PR}$.

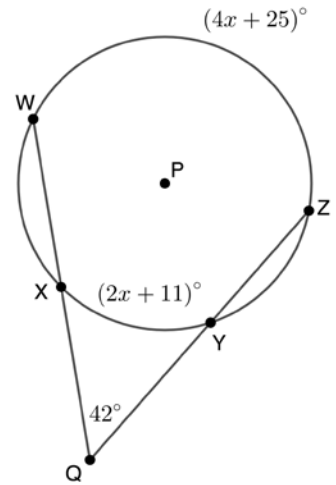
5. Find the $m\widehat{ST}$.

For questions 6-8, Circle P has points W , X , Y , and Z on the circle. Secant lines $\overline{WQ}$ and $\overline{ZQ}$ intersect at point Q outside the circle. The $m\widehat{WZ} = (4x + 25)^\circ$, $m\widehat{XY} = (2x + 11)^\circ$, and $m\angle XQY = 42^\circ$.

6. Find the value of x .

7. Find the $m\widehat{XY}$.

8. Find the $m\widehat{WZ}$.



For questions 9-10, Circle B has points R , L , Q , and X on the circle. Secant lines $\overline{XH}$ and $\overline{RH}$ intersect at point H outside the circle. The $m\widehat{RX} = 101^\circ$, $m\widehat{LQ} = 29^\circ$, and $m\angle LHQ = x^\circ$.

9. Write a formula that can be used to find the value of x .

10. Find the value of x .

